TECHNICAL SPECIFICATION · INTERNAL

ADEM Multi-Role Platform

Developer specification: architecture, roles, data model, APIs, flows, work plan, blockers and open decisions.

PRODUCT
ADEM: AI-assisted Digital Examination Management & Evaluation

DOCUMENT
Developer Specification v1.0

AUDIENCE
Backend, frontend and AI engineers

BOARD / SCOPE
CBSE · Class 10 and Class 12

STATUS
Agreed design, ready for build

DATE
30 October 2026

Confidential. For the internal engineering team only.

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01 Summary and agreed decisions

ADEM today is a single-role app: teachers upload a question paper, answer key and answer sheets, and AI grades them. This project turns it into a multi-role platform with four portals, a credit-based billing system and, in a separate track, automatic question paper generation using RAG.

OPERATOR
Platform Admin

A separate app. Manages every school, parent, credit, payment, dispute and audit record. Admin roles: Super Admin, Admin, Viewer.

TENANT OWNER
School Manager

One login per school. Invites teachers. Sees every exam, mark, answer sheet and report in the school, read-only. Buys credits.

SCHOOL STAFF
Teacher

The existing app, unchanged in behaviour. Now invited by the School Manager, scoped to one school, and evaluation uses school credits.

PUBLIC
Parent

Signs up publicly, not linked to any school. Adds several children, generates papers, the child sits a timed exam on screen, the parent uploads the sheet, and AI grades it.

Agreed decisions

Topic	Decision
Tenancy	One school = one organisation (no branches). Isolation by `school_id` (and `parent_id`) columns in one SQL database.
School onboarding	Public self-signup with email verification. The school stays **pending** until Platform Admin approves it.
Teacher onboarding	Invited by email by the School Manager. The teacher sets a password or uses Google with the invited email.
Google Sign-In	Every role. Creates accounts only for **schools and parents** (signup). For teachers and admins it only signs in to existing or invited accounts.
Accounts	One email = one role. No email holds two roles.
School Manager access	Full view of all school data, read-only. Past exams of deactivated teachers stay visible.
Parents	Several children per parent. Each child has a standard (10/12) and subjects.
Billing	Credits, not subscriptions. One wallet per school and per parent. Packs bought via Razorpay.
Parent credit charging	Two stages: a **fixed** charge when a paper is selected, then a larger **per-page** charge when the sheet is submitted for evaluation (refunded on failure).

Topic	Decision
Parent exam	"Start exam" button; paper on screen only; server-held timer with no pause; closes at time end; 1-hour upload window (image/PDF); report shows score and graded sheet; key never shown; retakes allowed.
Question papers	Generated live by RAG from subject, standard, lesson/unit. No human review before use, so automatic validation is mandatory.
Admin frontend	Separate React app. School, Teacher and Parent share the main web app.
Existing data	Existing teacher accounts and data are deleted after a full backup.
Tamil accuracy	Must be completed in this programme, in parallel with the platform work.

02 Current system baseline

Component	Today
Backend	FastAPI + SQLAlchemy, `backend/app`, port 8000. SQLite (`data/adem.db`, WAL, FKs on), or PostgreSQL via `database_url`.
Web frontend	React 18 + TypeScript + Vite + Tailwind, `frontend/`, port 5173, proxies `/api` to 8000.
Mobile	Expo app in `mobile/`.
AI: English and other subjects	InternVL 3.5 4B via Ollama reads documents and suggests marks.
AI: Tamil	IndicOCR service (`indic_ocr_service/`, port 8090) reads; InternVL grades. The answer key is the authority.
Subject formats	`backend/app/cbse/class10/subjects/*/format.json` (10 subjects) and `class12/subjects/tamil`. Shared rules in `app/cbse/common/`.
Auth	Teacher-only: `POST /auth/login`, `GET /auth/me`; JWT (HS256, `sub` = teacher id, 480 min). Teachers are created by script (`app/create_teacher.py`).
Ownership model	Every class, subject, exam, school profile and report template hangs off `teacher_id`. **There is no School entity.**
Email, payments	None.

> **Main structural gap**
>
> The backbone change is introducing `School` and a role-aware `User`, and rescoping every existing query from "this teacher" to "this teacher, inside this school". The `current_teacher` dependency and `deps.get_exam` are the choke points.

03 Target architecture

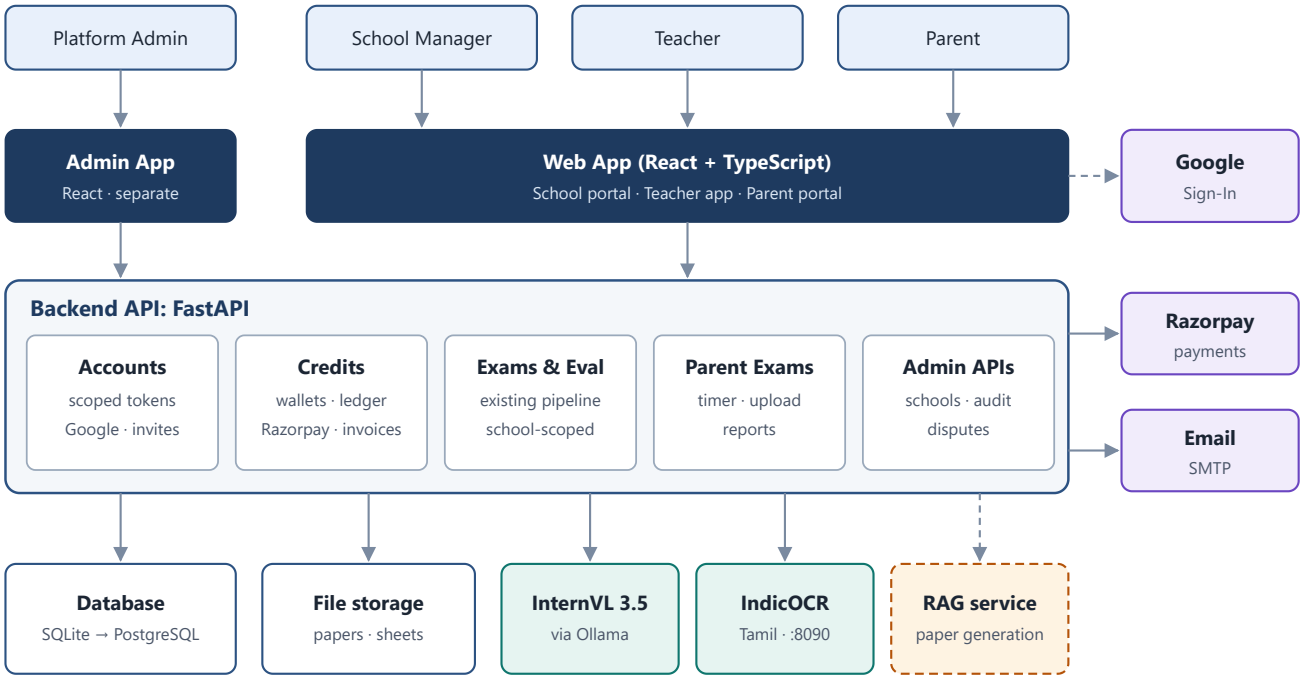


Figure 1. Target architecture. Dashed amber = new service. Purple = external services.

Architecture principles

- **One backend, one database.** All portals call the same FastAPI app. Tenancy is enforced by row scoping (`school_id`, `parent_id`), never trusted from the client.
- **Token scope per portal.** Each token carries a `scope` (`admin`, `school`, `teacher`, `parent`). An admin token cannot call parent routes and vice versa.
- **Role and status re-read on every request.** The token identifies the user; role, status and `token_version` are read from the database each call. Revocation and role changes take effect immediately.
- **Evaluation pipeline untouched.** Parent exams and RAG papers are written into the existing `Exam`, `ExamQuestion` and `AnswerKey` structures, so grading runs exactly as today.
- **RAG as a separate service.** Heavy indexing and generation jobs must not compete with live grading in the backend process. The backend calls it over HTTP. **recommended, to confirm**
- **Money is append-only.** Credits move only through ledger rows; balances are derived and updated in the same transaction.

04 Roles and permissions

Capability	Platform Admin	School Manager	Teacher	Parent
Self-signup	–	✓	invite	✓
Google Sign-In	**login**	**signup + login**	**login (invited email)**	**signup + login**
Manage schools / parents	✓	–	–	–
Approve / suspend schools	✓	–	–	–
Invite / deactivate teachers	**✓ (support)**	✓	–	–
Create exams, upload, grade, finalise	–	–	✓	–
View all school exams, marks, sheets	✓	**✓ read-only**	own only	–
Children, papers, timed exams	–	–	–	✓
Buy credits	–	**✓ school wallet**	–	**✓ own wallet**
Grant / deduct credits	**✓ audited**	–	–	–
Raise grade dispute	–	✓	✓	✓
Resolve disputes, refunds	✓	–	–	–
Login as another user (support)	**Super Admin, audited**	–	–	–

Admin console roles

Role	Permissions
Super Admin	Everything, including console users, impersonation, deletion of schools/parents.
Admin	Schools, parents, credits, payments, disputes, support. No console user management, no impersonation.
Viewer	Read every screen; every write route returns 403.

An unknown or missing role always resolves to the least privileged role (Viewer), never to Super Admin.

05 Accounts, authentication and signup

Session tokens

- JWT HS256 with claims `sub` (user id), `scope`, `tv` (token version), `exp`. The existing secret handling in `config.py` is reused.
- The guard per scope (`require_admin`, `require_school_manager`, `require_teacher`, `require_parent`) checks scope, loads the user, and rejects if status ≠ active, role mismatches, `tv` ≠ `token_version`, or the school is not active.
- "Sign out all sessions", a password change or a reset increments `token_version`.
- While `must_change_password` is set, only `/auth/me`, `/auth/change-password` and `/auth/logout` are allowed.

Password security

- bcrypt (existing `hash_password`). Minimum length 8 with a basic strength rule.
- Lockout: 5 failures → 15-minute lock per account and per IP. Admin can unlock.
- Rate limits: login, signup, forgot-password and verify-email per IP.
- Reset and invite links: random token, only its SHA-256 hash stored, single use, expiry (reset 1 h, invite 7 days, signup verify 24 h).
- Responses never reveal whether an email exists (forgot-password, signup).

Google Sign-In

1. The frontend uses Google Identity Services and receives an ID token.
2. `POST /api/auth/google` with `{id_token, portal}`. The backend verifies signature, `aud` = our Client ID, `iss`, `exp`, and `email_verified = true`.
3. Matching: first by `google_sub`, then by email. On the first email match, `google_sub` is linked to the account.
4. The portal must match the account's role. A parent's Google account cannot open the school portal.
5. No account found: the **school** and **parent** portals continue to signup (email already verified, verification link skipped). The **teacher** and **admin** portals return "No account for this email".
6. A Google-only account can set a password later through "Forgot password".

Signup and onboarding flows

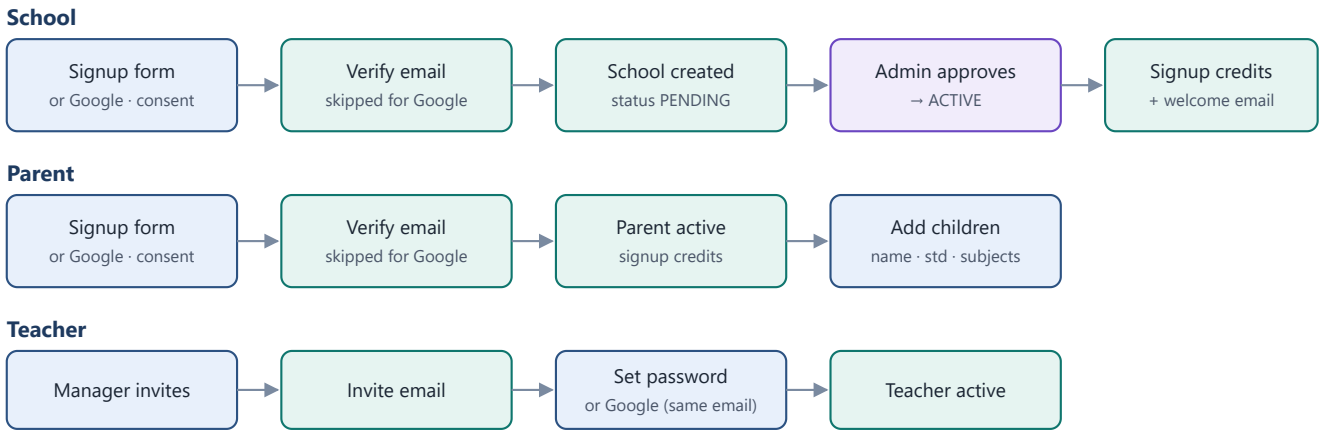


Figure 2. Onboarding flows. Blue = user action, teal = system, purple = Platform Admin.

Abuse controls

- Signup credits are granted once per email, per phone (last 10 digits) and per school, and to schools only after approval.
- The signup form answers the same whether or not the email exists. An existing account gets a "you already have an account" email instead.
- Consent records (privacy policy and terms version, time, IP) are stored at signup. Parents explicitly consent to processing their children's data.

06 Feature specification by portal

Tags: NEW new build · KEEP existing, re-scoped · RAG depends on RAG track.

6.1 Admin App (separate)

Area	Features
Auth NEW	Email + password, Google (existing accounts), forgot/reset, forced change, lockout + unlock, logout, sign out all sessions, session expiry.
Dashboard	Totals (schools, parents, children, teachers); evaluations today/month by subject; credits sold vs used; revenue; attention list (pending schools, failed payments, failed evaluations, low-credit schools, open disputes); AI status (GPU queue, InternVL, IndicOCR, RAG).
Schools	List, search, filter; detail (profile, manager, teachers, classes/students, exams, wallet + ledger); approve; suspend/activate; edit, change manager email; reset password, unlock; login as (Super Admin, audited); delete.
Parents	List, search; detail (children, exams, results, wallet + ledger); suspend/activate; reset, unlock; login as (audited); delete.
Credits	Credit packs CRUD; cost settings (paper fixed, per page, subject multipliers); signup credit amounts; manual grant/deduct with reason; global ledger.
Payments	Razorpay orders (created/paid/failed/refunded); detail; refund; GST invoices; gateway settings (live/simulation).
Evaluations monitor	Running, queued and failed evaluations across all tenants; retry (held credits released on failure).
Generated papers RAG	Log of every generated paper and key with sources and validation results; spot-check; disable a paper.
Knowledge base RAG	Upload textbooks, syllabus, sample papers, marking schemes per class/subject; index status; re-index.
Disputes and support	Grade disputes (review, resolve, refund credits); support tickets (reply, close).
Audit log	Every write: actor, role, target, school, IP, time. Filter and export.
Console users	Invite, change role (Super Admin / Admin / Viewer), revoke, reset, unlock.
Settings	Email (SMTP) and test send; Google Client ID; GST and invoice details; platform settings.

6.2 School portal

Area	Features
Signup / auth NEW	Public signup (school + manager details, password or Google, consent); verify email; pending-approval screen; login; forgot/reset/change; lockout; logout.
Dashboard	Exams this term; evaluation progress; class averages, pass rates, toppers; teacher activity; credit balance with low-credit warning.

Area	Features
Teachers	Invite by email, resend invite, deactivate/reactivate, reset password, unlock.
Exams (read-only)	All exams across teachers, filtered by class/subject/teacher/status; paper and key; every student's marks; answer sheets with per-question grading; results, marksheets, report cards.
Classes and students	View all.
Reports	School-wide performance; class/subject/teacher comparisons; Excel/PDF export.
School profile KEEP	Name, address, principal, logo; report and report card templates (moved from teacher level to school level).
Credits	Balance; buy pack (Razorpay); ledger (which exam, sheet and teacher used credits); payment history; GST invoices.
Academic year	Year rollover: promote students, archive past years with results still viewable.
Disputes, support	Raise grade dispute; contact support.

6.3 Teacher app

Area	Features
Auth NEW	Accept invite and set password; login with password or Google (invited email only); forgot/reset/change; lockout; logout.
Existing workflow KEEP	Dashboard; classes, subjects, students (add, edit, import, export); create exam; upload paper and key; upload answer sheets; AI evaluation; review and correct; finalise; results, marksheets, report cards.
Credits NEW	Cost quote before evaluation (pages × rate); school wallet is charged; clear "insufficient credits, ask your school manager" state. Re-grading after a teacher correction is free.

6.4 Parent portal

Area	Features
Signup / auth NEW	Public signup (password or Google, consent incl. children's data); verify email; login; forgot/reset/change; lockout; logout.
Children	Add, edit, remove (name, standard 10/12, subjects); switch child.
Dashboard	Per child: recent exams and scores, progress over time, weak chapters; credit balance.
Exam RAG	Choose subject, standard, lesson/unit → paper generated (fixed charge) → Start exam → on-screen paper with server timer → auto-close → 1-hour upload (images/PDF) → quality check → cost confirm → evaluation. See section 07.
Reports	Score and graded answer sheet (key not shown); strengths and weak chapters; subject progress; download report PDF.
Retakes	Take another paper on the same units (new paper, new charge).

Area	Features
Credits	Balance; buy pack (Razorpay); ledger; payment history; invoices.
Disputes, support	Flag a graded question; contact support; delete my data.

6.5 Shared platform features

- **Email notifications:** verification, invites, password reset, school approved, purchase receipt, evaluation complete, low credits, upload window closing (parent).
- **Upload quality check** before any charge: page count, blur (variance of Laplacian), orientation, darkness, unsupported or corrupt files. Fail → ask the user to retake.
- **Mobile-friendly** Parent and School portals; camera capture for page-by-page upload.
- **Data rights:** consent records; "delete my data"; existing post-finalisation file purge stays.

07 Parent exam flow

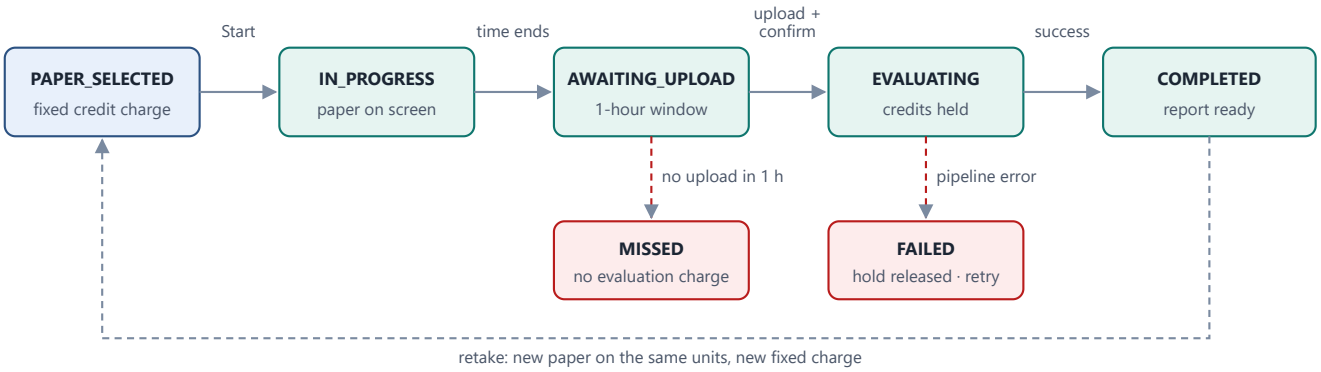


Figure 3. Parent exam state machine. All transitions and times are decided by the server.

Step	Rules
Select paper	Parent chooses child, subject, standard and lesson(s)/unit(s). Balance must cover the fixed charge. RAG generates the paper and key synchronously, or as a short job with a progress state. Ledger `PAPER_CHARGE`. The key is stored locked and never sent to parent endpoints.
Start exam	Explicit button. Server sets `started_at`, `ends_at = started_at + duration`, `upload_closes_at = ends_at + 60 min`.
Paper display	Pages served as server-rendered images (no PDF, no download). Each page is watermarked with the parent email and exam id. The endpoint refuses after `ends_at`. The client countdown is cosmetic and always re-syncs to server time. Closing the browser does not pause the exam. No pause.
Upload	Images (JPG/PNG/HEIC) or PDF, multi-page, reorder before submit. Quality check per page. Rejected after `upload_closes_at` → MISSED.
Confirm	Quote = pages × per-page rate × subject multiplier. Insufficient balance → buy credits (the upload window keeps running). Confirm → `EVAL_HOLD`.
Evaluate	Existing pipeline (InternVL; IndicOCR + InternVL for Tamil) against the locked key. Success → `EVAL_CAPTURE`. Failure → `EVAL_RELEASE`, retry allowed.
Report	Score, graded answer sheet with per-question marks; strengths and weak chapters; PDF download. The key and model answers are not shown.

> **Note on exam integrity**
>
> Screen-only display blocks download and print, not phone photos of the screen. The watermark discourages sharing. The product should not claim the exam is cheat-proof.

08 Credit system

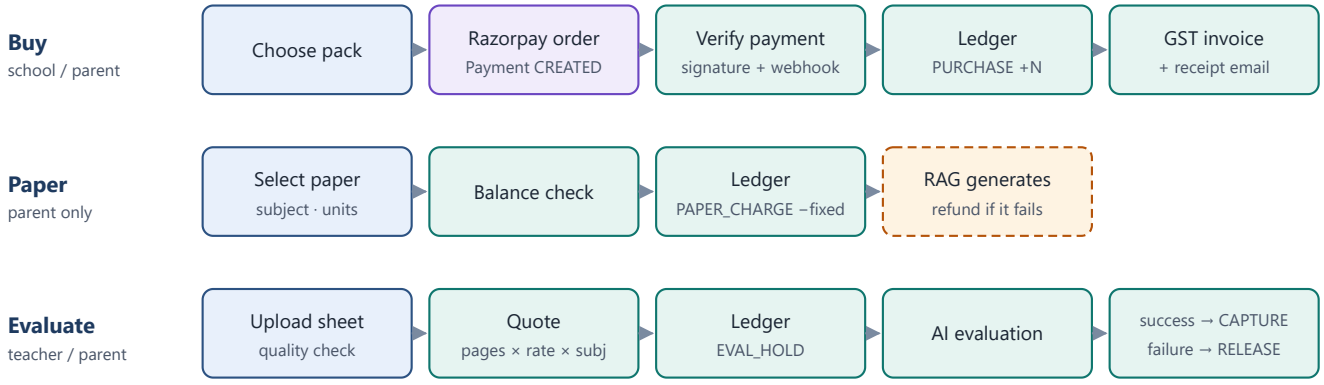


Figure 4. Credit movements. Every movement is one append-only ledger row.

Rules

- **Wallets:** one per school (shared by its teachers) and one per parent. Fields `balance` (available) and `held`.
- **Ledger types:** `PURCHASE`, `SIGNUP_BONUS`, `ADMIN_GRANT`, `ADMIN_DEDUCT`, `PAPER_CHARGE`, `PAPER_REFUND`, `EVAL_HOLD`, `EVAL_CAPTURE`, `EVAL_RELEASE`, `DISPUTE_REFUND`, `PAYMENT_REFUND`, `EXPIRY`.
- **Atomicity:** the ledger row and the wallet update happen in one transaction (`SELECT … FOR UPDATE` on PostgreSQL; serialized writes on SQLite). Balance can never go negative.
- **Idempotency:** Razorpay payment ids and evaluation ids are unique ledger references, so webhook retries or double clicks never double-credit or double-charge.
- **Free re-grading** after a teacher correction. Disputes upheld by Admin refund the affected credits.
- **Zero balance:** new paper or evaluation is blocked with a "buy credits" prompt **to confirm**.

Configurable values **pending business input**

Setting	Placeholder used until confirmed
Credit packs (credits / price ₹)	100 / 500 / 1000 credits; prices to be set
Paper selection (fixed)	5 credits
Evaluation per page	2 credits
Subject multiplier (Tamil)	1.5×
Signup credits (school / parent)	50 / 20
Expiry	12 months from purchase

09 Question paper generation (RAG)

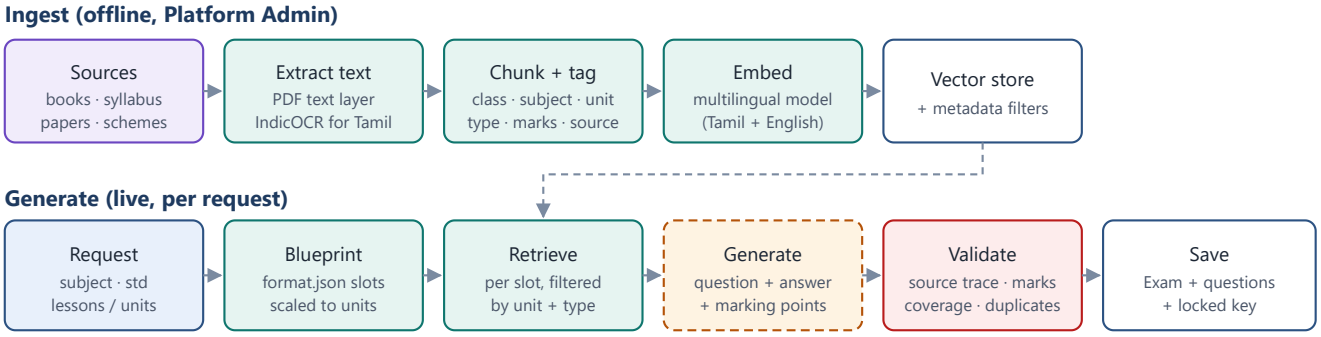


Figure 5. RAG pipeline. Validation failures regenerate the failing question only.

Design

- **Scope:** CBSE Class 10 and 12. Start with a pilot of one or two subjects that already have sample papers and marking schemes, then widen.
- **Chunk metadata:** class, subject, chapter, unit, topic, content type (theory, example, exercise, past question), question type, marks, difficulty, source document and page. Metadata filters come first, similarity second.
- **Blueprint:** the subject's `format.json` defines sections, question types and marks. For unit-level papers it is scaled down proportionally, keeping the mix of question types.
- **Output contract:** structured JSON per question (number, section, type, marks, text, options, model answer, marking points, source refs). It is written into `ExamQuestion` and `AnswerKey`, so the evaluation pipeline needs no change.
- **Validation (mandatory, since papers are not reviewed by a human before use):**
  - every key answer must be supported by a retrieved source passage, otherwise regenerate;
  - the marks total equals the target;
  - all questions are within the selected units;
  - no duplicate or near-duplicate questions, and no repeat of the child's recent papers;
  - every question has marking points summing to its marks;
  - MCQs have exactly one correct option.
- **Paper rendering:** HTML → page images for screen-only display. Maths uses KaTeX (already a frontend dependency).
- **Observability:** store prompts, retrieved chunks, model, validation results and latency per paper, for Admin spot checks and dispute resolution.

> **Quality gate**
>
> Parent exams go live only after the pilot passes a quality review: a sample of generated papers is checked by a subject expert for correctness of keys, syllabus fit and CBSE style. If it fails, the parent portal ships with paper generation shown as "coming soon".

10 Data model

New tables

Table	Key columns
`schools`	id, name, board, address, city, state, phone, principal_name, logo_file, status (PENDING | ACTIVE | SUSPENDED), approved_at, approved_by, created_at
`users`	id, email (unique), name, phone, role (platform_admin | school_manager | teacher | parent), admin_role (super_admin | admin | viewer), school_id (null for admin and parent), status (invited | active | suspended | removed), password_hash (nullable), google_sub (unique, nullable), email_verified_at, must_change_password, failed_attempts, locked_until, token_version, last_login_at, created_by, created_at
`auth_tokens`	id, user_id, kind (invite | reset | verify), token_hash, expires_at, used_at
`signup_requests`	id, kind (school | parent), email, phone_key, payload (JSON), via_google, status, created_user_id, created_at
`consents`	id, user_id, kind (privacy | terms | child_data), version, accepted_at, ip
`children`	id, parent_id → users, name, standard (10 | 12), subjects (JSON), created_at
`wallets`	id, owner_type (school | parent), owner_id, balance, held, updated_at
`credit_ledger`	id, wallet_id, type, amount (±), balance_after, ref_type, ref_id (unique per type), note, actor_id, created_at (append-only)
`credit_packs`	id, name, credits, price_inr, active, sort_order
`credit_settings`	key, value (paper_fixed, eval_per_page, subject multipliers, signup amounts, expiry_days)
`payments`	id, wallet_id, pack_id, razorpay_order_id, razorpay_payment_id (unique), signature, amount_inr, gst_amount, status (CREATED | PAID | FAILED | REFUNDED), invoice_no, created_at, paid_at
`generated_papers`	id, standard, subject, units (JSON), blueprint, exam_id → exams, sources (JSON), validation (JSON), model, latency_ms, created_at
`parent_exams`	id, child_id, paper_id, status, selected_at, started_at, ends_at, upload_closes_at, uploaded_at, pages, evaluation_id, credits_paper, credits_eval
`disputes`	id, raised_by, evaluation_id, question_id, reason, status, resolution, refund_credits, resolved_by, resolved_at
`support_tickets`	id, user_id, subject, message, status, reply, replied_by, created_at

Changes to existing tables

Table	Change
`teachers`	Replaced by `users` with role = teacher. `teacher_id` foreign keys point to `users.id`.

Table	Change
`classes`, `subjects`, `students`, `exams`	Add `school_id` (indexed, not null for school data). Every query filters by it.
`exams`	Add `origin` (TEACHER | PARENT | GENERATED) and nullable `owner_parent_id`, so parent exams reuse the pipeline but stay invisible to schools.
`evaluations`	Add `credits_charged`, `ledger_ref`, `pages`.
`school_profiles`, `report_templates`	Keyed by `school_id` instead of `teacher_id`.
`audit_logs`	Add actor_id, actor_role, school_id, target_type, target_id, ip.

> **Migration**
>
> Existing teacher accounts and data are deleted, not migrated. Take and verify a full backup of the database and the `data/` folder first. The first Super Admin is seeded by a script, replacing `create_teacher.py`.

11 API surface

All routes are under `/api`. Scope column = required token scope.

Scope	Endpoints
public	`POST /auth/login` {email, password, portal} · `POST /auth/google` {id_token, portal} · `POST /auth/forgot-password` · `POST /auth/reset-password` · `POST /auth/invite/accept` · `GET /public/credit-packs` · `POST /public/signup/school` · `POST /public/signup/parent` · `POST /public/signup/verify-email` · `POST /public/razorpay/webhook`
any signed in	`GET /auth/me` · `POST /auth/change-password` · `POST /auth/logout` · `POST /auth/logout-all` · `POST /support` · `POST /disputes`
admin	`GET /admin/dashboard` · `GET|POST /admin/schools` · `GET|PUT|DELETE /admin/schools/{id}` · `POST /admin/schools/{id}/approve|status|reset-password|unlock|impersonate` · `GET /admin/parents` · `GET|DELETE /admin/parents/{id}` · `POST /admin/parents/{id}/status|reset-password|unlock|impersonate` · `CRUD /admin/credit-packs` · `GET|PUT /admin/credit-settings` · `POST /admin/wallets/{id}/adjust` · `GET /admin/ledger` · `GET /admin/payments` · `POST /admin/payments/{id}/refund` · `GET /admin/evaluations` · `POST /admin/evaluations/{id}/retry` · `GET /admin/papers` · `POST /admin/knowledge/upload` · `GET /admin/knowledge` · `GET /admin/disputes` · `POST /admin/disputes/{id}/resolve` · `GET /admin/support` · `POST /admin/support/{id}/reply` · `GET /admin/audit-logs` · `CRUD /admin/users` · `GET|PUT /admin/settings`
school	`GET /school/dashboard` · `GET|POST /school/teachers` · `PATCH /school/teachers/{id}` · `POST /school/teachers/{id}/resend-invite|reset-password|unlock` · `GET /school/exams` · `GET /school/exams/{id}` · `GET /school/exams/{id}/results` · `GET /school/evaluations/{id}` · `GET /school/classes` · `GET /school/students` · `GET /school/reports/*` · `GET|PUT /school/profile` · report templates (moved from today's school routes) · `GET /school/credits` · `POST /school/credits/order|verify` · `GET /school/invoices/{id}` · `POST /school/year-rollover`
teacher	All existing routes unchanged in shape, now scoped by `school_id`. Adds `GET /evaluations/{id}/quote`; evaluation start places a credit hold.
parent	`CRUD /parent/children` · `GET /parent/dashboard` · `POST /parent/papers` {child_id, subject, standard, units} · `GET /parent/exams` · `POST /parent/exams/{id}/start` · `GET /parent/exams/{id}` (state + server time) · `GET /parent/exams/{id}/pages/{n}` (only while IN_PROGRESS) · `POST /parent/exams/{id}/upload` · `GET /parent/exams/{id}/quote` · `POST /parent/exams/{id}/evaluate` · `GET /parent/exams/{id}/report` · `GET /parent/credits` · `POST /parent/credits/order|verify` · `POST /parent/delete-my-data`

12 Frontend structure

App	Structure
Admin App NEW	Separate Vite + React + Tailwind project (e.g. `admin/`), own dev port. Sidebar: Dashboard, Schools, Parents, Credits, Payments, Evaluations, Papers, Knowledge base, Disputes, Support, Audit log, Users, Settings. Shared table, drawer and form components.
Web App	Existing `frontend/`. Route areas: `/school/*`, `/teacher/*` (today's pages), `/parent/*`, plus public `/login`, `/signup/school`, `/signup/parent`, `/verify`, `/reset`, `/invite`. The auth context stores scope and role; route guards per area; layouts per role.
Shared UI	Login with "Continue with Google", password strength, lockout messages, credit balance chip, buy-credits modal (Razorpay Checkout), ledger table, upload with camera capture and page reorder.
Parent exam screen	Full-screen paper viewer (images, zoom, page nav), server-synced countdown, auto-close, upload step, quote and confirm, evaluation progress, report view.

13 Tamil accuracy track

This runs in parallel with the platform work and must be completed in this programme. Principles stay as they are: IndicOCR reads, InternVL grades, the answer key is the authority, InternVL judges meaning per marking point, and the rubric is reviewed once. Essays and letters stay unchanged until their rubrics are supplied.

Workstream	Output
Replay harness	Re-run stored Tamil evaluations offline against a fixed reference set; per-question diff of marks vs the teacher's final marks.
Self-check	Automatic consistency checks of read paper + key + sheet (question numbering, mark totals, missing answers) before grading.
Alignment	Robust mapping of handwritten answers to question ids (numbering variants, continuation pages).
Fixes and measurement	Iterate on the largest error classes; report accuracy before and after on the reference set.

> **GPU sharing**
>
> Tamil grading, English grading and RAG generation share GPU capacity. Keep the existing queue switching rules. RAG generation must not pre-empt live grading.

14 Work plan

Two tracks run in parallel. Phases are ordered by dependency. Each phase ends with a demo-able result.

Track A: Platform

Phase	Goal	Tasks
A1	Foundation	Backup and verified wipe of existing data; new tables and model changes; Super Admin seed script; scoped tokens, per-scope guards, `token_version`, lockout, forced password change, rate limits; replace `current_teacher`; add `school_id` scoping to every teacher route and `deps.get_exam`.
A2	Accounts	Email service and templates; invites; forgot/reset; Google Sign-In with linking rules; public school and parent signup; verify-email; consent records; school approval; signup credit abuse limits.
A3	Credits and payments	Wallets, ledger, packs, settings; hold/capture/release; paper charge; Razorpay order, verify and webhook (simulation mode first); GST invoice PDF; receipt emails.
A4	Admin App	Scaffold and login; dashboard; schools (approve, suspend, impersonate); parents; credits; payments; evaluations monitor; disputes; support; audit; console users; settings.
A5	School portal + Teacher	School dashboard, teachers, read-only exams with full drill-down, reports, profile/templates move, credits; teacher invite acceptance and Google; evaluation quote and charge; full regression of the existing teacher workflow.
A6	Parent portal	Children, dashboard, credits; exam runner (start, server timer, screen-only pages with watermark, auto-close); upload with quality check; quote and confirm; evaluation; report; retakes; disputes; notifications. Uses sample papers until RAG passes its gate.
A7	Hardening and release	End-to-end tests (section 15); security review; mobile pass; academic year rollover; delete-my-data; production configuration (domain, HTTPS, Google origins, Razorpay live keys, SMTP); deployment; demo data.

Track B: AI

Phase	Goal	Tasks
B1	Tamil baseline	Replay harness and reference set; baseline accuracy.
B2	Tamil fixes	Self-check, alignment, fixes on top error classes; re-measure.
B3	RAG service	Service skeleton; ingest and extraction (IndicOCR for Tamil); chunking with metadata; embeddings; vector store.
B4	RAG generation	Blueprint from `format.json`; retrieval per slot; generation; validation; write into exam structures; page rendering.

Phase	Goal	Tasks
Gate	Quality gate	Expert review of a sample of pilot papers and keys. Pass → enable for parents. Fail → parent paper generation stays "coming soon".
B5	RAG integration	Parent paper endpoint; logging for Admin; GPU queue integration; widen subjects.

Release scope

Release 1 (must ship)

- All logins, signup, Google, invites, resets
- Admin: schools, parents, credits, payments, audit, users
- School Manager full read-only view, teachers, credits
- Teacher app on new accounts with credits
- Parent flow end to end, reports
- Wallets, ledger, Razorpay, invoices
- Tamil accuracy improvements
- Consent, approval, quality check, disputes

Release 2 (follow-up)

- School-wide comparison reports and exports
- Detailed analytics dashboards
- Academic year rollover (if not reached in A7)
- RAG beyond pilot subjects
- Report template designer at school level

15 Testing and definition of done

Automated end-to-end suites

Suite	Must prove
Tenant isolation	School A cannot read or write anything of school B; a parent cannot see another parent's children or exams; schools never see parent exams.
Roles and scopes	Every route rejects the wrong scope and role; Viewer cannot write; a suspended user or school is rejected on the next request.
Sessions	Logout-all, password change and reset invalidate old tokens; lockout and unlock; forced password change limits.
Google	Linking by email, wrong portal rejected, no account creation for teacher/admin portals, `email_verified` required.
Signup	Verify-email expiry and single use; same response for existing emails; one signup credit per email, phone and school; school pending until approved.
Credits	Hold/capture/release; concurrent charges never go negative; webhook replay never double-credits; refunds on failure and disputes.
Parent exam timing	Pages refused after `ends_at`; upload refused after the window closes; refresh does not reset the timer; MISSED charges nothing more.
Evaluation regression	The existing teacher flow (English and Tamil) gives identical results before and after the re-scoping.
RAG validation	Marks total, unit coverage, duplicates, key source-trace and MCQ checks are enforced.

Definition of done (per feature)

- Backend route with scope guard, tenancy scoping, validation and audit log entry.
- UI states: loading, empty, error, success; mobile layout for the Parent and School portals.
- Covered by the relevant e2e suite; typecheck passes (`npm run typecheck`).
- No secrets in code; configuration through environment and settings.

A Appendix A: Complete feature map

Every feature of the platform as a tree, grouped by portal and shared system. Use it as the checklist for Release 1 and Release 2.

```
ADEM Platform
│
├── 1. ADMIN APP  (separate app · Platform Admin)
│   ├── Auth
│   │   ├── Login: email + password
│   │   ├── Login: Google (existing admin accounts only)
│   │   ├── Forgot / reset password (email link)
│   │   ├── Change password (forced after invite or reset)
│   │   ├── Lockout after repeated failures + unlock
│   │   ├── Logout / sign out all sessions
│   │   └── Session expiry
│   ├── Dashboard
│   │   ├── Totals: schools, parents, children, teachers
│   │   ├── Evaluations today / this month, by subject
│   │   ├── Credits sold, credits used, revenue
│   │   ├── Attention: pending schools, failed payments, failed evaluations,
│   │   │   low-credit schools, open disputes
│   │   └── AI status: GPU queue, InternVL, IndicOCR, paper generator
│   ├── Schools
│   │   ├── List (search, filter by status / credits)
│   │   ├── School details
│   │   │   ├── Profile (name, logo, address, principal)
│   │   │   ├── Manager account
│   │   │   ├── Teachers
│   │   │   ├── Classes / students
│   │   │   ├── Exams + evaluations
│   │   │   └── Credit wallet + ledger
│   │   ├── Approve new school
│   │   ├── Activate / suspend school
│   │   ├── Edit school / change manager email
│   │   ├── Reset manager password / unlock
│   │   ├── Login as school (support, audited)
│   │   └── Delete school
│   ├── Parents
│   │   ├── List (search, filter)
│   │   ├── Parent details
│   │   │   ├── Children
│   │   │   ├── Exams + results
│   │   │   └── Credit wallet + ledger
│   │   ├── Suspend / activate
│   │   ├── Reset password / unlock
│   │   ├── Login as parent (support, audited)
│   │   └── Delete parent
│   ├── Credits
│   │   ├── Credit packs (credits, price, discount, active)
│   │   ├── Credit costs (paper fixed charge, evaluation per page, subject rates)
│   │   ├── Signup free credits (one per email / phone / school)
│   │   ├── Manual grant / deduct (with reason, audited)
│   │   └── Global ledger (every credit movement)
│   ├── Payments
│   │   ├── Razorpay orders: created / paid / failed / refunded
│   │   ├── Payment details + refund
│   │   ├── GST invoices
│   │   └── Gateway settings (live / simulation)
│   ├── Question Papers  [paper generator]
│   │   ├── QP requests inbox (from schools)
│   │   ├── Generated papers log (paper, key, sources, checks)
```

```
│   │   ├── Spot-check / disable a paper
│   │   └── Knowledge base (upload books, syllabus, sample papers, marking schemes)
│   ├── Evaluations monitor
│   │   ├── Running / queued / failed evaluations (all users)
│   │   └── Retry failed (held credits released)
│   ├── Disputes (review grade, resolve, refund credits)
│   ├── Support tickets (reply, close)
│   ├── Audit log (who, what, when, IP)
│   ├── Console users
│   │   ├── Roles: Super Admin / Admin / Viewer
│   │   ├── Invite, change role, revoke
│   │   └── Reset password / unlock
│   └── Settings
│       ├── Email (SMTP) + test send
│       ├── Google sign-in client
│       ├── GST + invoice details
│       └── Platform settings
│
├── 2. SCHOOL MANAGEMENT  (one login per school · full view · read-only)
│   ├── Signup (public)
│   │   ├── School details + manager details + password
│   │   ├── Or "Continue with Google"
│   │   ├── Consent (privacy policy, terms)
│   │   ├── Email verification (skipped for Google)
│   │   ├── Pending until Platform Admin approves
│   │   └── Free signup credits after approval
│   ├── Auth
│   │   ├── Login: email + password / Google
│   │   ├── Forgot / reset / change password
│   │   ├── Lockout + logout
│   │   └── Session expiry
│   ├── Dashboard
│   │   ├── Exams this term, evaluation progress
│   │   ├── Class averages, pass rates, toppers
│   │   ├── Teacher activity
│   │   └── Credit balance + low-credit warning
│   ├── Teachers
│   │   ├── Invite by email (teacher sets password or uses Google)
│   │   ├── Resend invite
│   │   ├── Deactivate / reactivate (past exams stay visible)
│   │   └── Reset password / unlock
│   ├── Exams (read-only, all teachers)
│   │   ├── Filter by class / subject / teacher / status
│   │   ├── Paper + answer key
│   │   ├── Every student's marks
│   │   ├── Answer sheets + per-question grading
│   │   ├── Results, marksheets
│   │   └── Report cards
│   ├── Classes & students (view)
│   ├── Reports
│   │   ├── School-wide performance
│   │   ├── Class / subject / teacher comparisons
│   │   └── Export (Excel / PDF)
│   ├── Question papers  [paper generator]
│   │   ├── Request a QP (class, subject, chapters, marks, difficulty, due date)
│   │   ├── Track request status
│   │   └── Received papers → assign to a teacher's exam
│   ├── School profile
│   │   ├── Name, address, principal, logo
│   │   └── Report / report card templates
│   ├── Academic year rollover (promote students, archive past years)
│   ├── Credits
│   │   ├── Balance
│   │   ├── Buy credit pack (Razorpay)
│   │   ├── Ledger (which exam, sheet, teacher used credits)
│   │   └── Payment history + GST invoices
```

```
│   └── Disputes + support
│
├── 3. TEACHER  (current app · invited by the school)
│   ├── Auth
│   │   ├── Accept invite → set password
│   │   ├── Login: email + password / Google (invited email only)
│   │   ├── Forgot / reset / change password
│   │   └── Lockout + logout
│   ├── Dashboard (as today)
│   ├── Classes, subjects, students (as today: add, edit, import, export)
│   ├── Exams (as today)
│   │   ├── Create exam
│   │   ├── Upload question paper + answer key
│   │   ├── Start from a school-received QP  [paper generator]
│   │   ├── Upload answer sheets
│   │   ├── Credit cost shown before evaluation (pages × rate)
│   │   ├── AI evaluation (uses school credits)
│   │   ├── Review / correct grading (re-grade free)
│   │   └── Finalize
│   ├── Results, marksheets, report cards (as today)
│   └── Disputes + support
│
├── 4. PARENT  (public · not linked to any school)
│   ├── Signup (public)
│   │   ├── Name, email, phone, password
│   │   ├── Or "Continue with Google"
│   │   ├── Consent (incl. processing children's data)
│   │   ├── Email verification (skipped for Google)
│   │   └── Free signup credits (one per email / phone)
│   ├── Auth
│   │   ├── Login: email + password / Google
│   │   ├── Forgot / reset / change password
│   │   └── Lockout + logout
│   ├── Children (several per parent)
│   │   ├── Add / edit / remove child (name, standard 10 / 12, subjects)
│   │   └── Switch child
│   ├── Dashboard (per child)
│   │   ├── Recent exams + scores
│   │   ├── Progress over time, weak chapters
│   │   └── Credit balance
│   ├── Exam
│   │   ├── Choose subject, standard, lesson / unit
│   │   ├── Paper generated automatically  [paper generator]
│   │   │   └── Fixed credit charge · answer key locked
│   │   ├── "Start exam" button
│   │   ├── Paper on screen only, countdown timer (no pause, no download / print)
│   │   ├── Time ends → paper closes
│   │   ├── Upload answer sheet within 1 hour (images or PDF)
│   │   ├── Photo quality check before any charge
│   │   ├── Confirm evaluation credits (per page)
│   │   └── AI evaluation
│   ├── Reports
│   │   ├── Score + graded answer sheet (key not shown)
│   │   ├── Strengths / weak chapters
│   │   ├── Subject-wise progress
│   │   └── Download report PDF
│   ├── Retake (new paper on the same units)
│   ├── Credits
│   │   ├── Balance
│   │   ├── Buy credit pack (Razorpay)
│   │   ├── Ledger
│   │   └── Payment history + invoices
│   ├── Disputes + support
│   └── Delete my data
│
├── 5. CREDIT SYSTEM  (shared)
```

```
│   ├── Wallets: one per school (shared by teachers), one per parent
│   ├── Packs bought via Razorpay → credits added + GST invoice
│   ├── Parent paper: fixed charge on selection
│   ├── Evaluation: per page · hold on start → charge on success → refund on failure
│   ├── Re-grade after teacher correction: free
│   ├── Upheld dispute: credits refunded
│   ├── Out of credits: blocked with "buy credits" prompt
│   ├── Admin manual grant / deduct
│   └── Ledger: every movement, visible to owner + Platform Admin
│
├── 6. EVALUATION ENGINE  (existing · shared by teachers + parents)
│   ├── English and other subjects: InternVL
│   ├── Tamil: IndicOCR reads → InternVL grades
│   ├── Answer key is the authority
│   ├── Tamil accuracy improvements
│   │   ├── Self-check against paper + key
│   │   ├── Replay harness
│   │   └── Alignment
│   └── GPU queue (current switching rules)
│
├── 7. QUESTION PAPER GENERATOR (RAG)  (CBSE · Class 10 & 12)
│   ├── Ingest: books, syllabus, sample papers, marking schemes
│   ├── Index: tagged chunks → embeddings → vector store
│   ├── Plan from the CBSE blueprint per subject
│   ├── Generate per question: question + model answer + marking points
│   ├── Automatic checks: source trace, marks total, coverage, duplicates
│   ├── Live for parents · on request for schools
│   ├── Quality gate (expert review of pilot papers before launch)
│   └── Output: on-screen paper + locked key, ready for AI evaluation
│
└── 8. PLATFORM  (shared)
    ├── Accounts: one email = one role
    ├── Google sign-in for every role
    │   (creates accounts only for schools + parents)
    ├── Email notifications: verification, invites, resets, receipts,
    │   school approved, evaluation complete, low credits, upload window closing
    ├── Data isolation per school and per parent
    ├── Audit log
    ├── Rate limits on login, signup, reset
    ├── Mobile-friendly School + Parent portals (camera upload)
    ├── Consent records + privacy policy + terms
    └── Fresh start: existing test data removed after backup
```

☐ Feature ☐ Added after review ☐ Depends on the paper generator

B Appendix B: Questions to clarify, blockers and issues

Everything that is still open, blocking or at risk, in one place. Review it at every phase hand-off.

B.1 Business questions to clarify

#	Question	Placeholder / recommendation until answered
1	Credit packs and prices	100 / 500 / 1000 credits, discount on larger packs
2	Credits for parent paper selection	5
3	Credits per evaluated page	2
4	Tamil multiplier	1.5×
5	Signup credits (school / parent)	50 (after approval) / 20
6	Credit expiry	12 months from purchase
7	Zero-balance behaviour	Block with buy-credits prompt
8	RAG pilot subjects	One or two Class 10 subjects with sample papers
9	Source books and documents for the pilot	Existing sample papers in `exam/` + NCERT books
10	Subject expert for the quality gate	Needed before parent launch
11	Fallback if RAG fails the gate	Parent paper generation shown as "coming soon"
12	Show answer key to parents after the report?	No (current decision)
13	Approve the five additions (GST invoices, notifications, year rollover, support form, mobile)	Built as approved
14	Privacy policy and terms text	Placeholders
15	Domain and hosting	Local / staging

B.2 Technical decisions to make

#	Decision	Recommendation
1	RAG service location	Separate service (like `indic_ocr_service`), called over HTTP.
2	RAG generation model	Evaluate local versus hosted on the pilot set for Tamil and English quality, latency and cost per paper.
3	Embedding model and vector store	A multilingual embedding model (e.g. bge-m3 / multilingual-e5); pgvector once on PostgreSQL, Chroma until then.
4	RAG pilot subjects	One or two Class 10 subjects with existing sample papers and marking schemes.
5	Credit values and expiry	See section 08 placeholders.

#	Decision	Recommendation
6	Zero-balance behaviour	Block with a buy-credits prompt.
7	Paper duration for unit-level papers	Scale the duration with the marks; show it before "Start exam".

B.3 Blockers and dependencies

Item	Owner	Needed by	Impact if missing
BLOCKER Database backup and approval to delete existing data	Product owner	A1	Phase A1 cannot start.
BLOCKER Uncommitted folder restructure in the working tree (`app/cbse/common`)	Tech lead	A1	New work gets mixed with unreviewed changes; it must be settled first.
BLOCKER Email provider (SMTP) and sender address/domain	Product owner	A2	No verification, invites or resets, so signup and onboarding are blocked.
Google OAuth Client ID + authorised origins	Product owner (in progress)	A2	No Google login; email login unaffected.
Privacy policy and terms text (DPDP Act 2023)	Product owner	A2	Consent screens without legal text.
Credit values (packs, prices, costs, signup credits, expiry)	Product owner	A3	Placeholders used.
Razorpay account, API keys, webhook secret	Product owner	A3 (live)	Simulation mode only, no real payments.
GST details (legal name, GSTIN, address)	Product owner	A3	Invoices without GST.
BLOCKER RAG generation model choice and GPU capacity	Tech lead	B3	RAG cannot be built, so parent paper generation is blocked.
RAG pilot subjects and source documents	Product owner	B3	Nothing to index.
RAG service location (separate service recommended)	Tech lead	B3	Skeleton cannot start.
Subject expert for the quality gate	Product owner	Gate	Parent exams cannot be enabled.
Production domain and hosting	Product owner	A7	Google origins, email links and payment callbacks cannot be configured.

B.4 Issues and risks

Risk	Level	Mitigation
Scope is large for the available time	HIGH	Release 1 / Release 2 split; parallel tracks; cut from Release 2 first, never from core flows.
Wrong answer keys in live generated papers	HIGH	Source-trace validation, quality gate, disputes with refunds, Admin spot checks, full generation logs.
GPU contention between grading and generation	MED	Generation queued behind live grading; keep queue rules; measure latency during A6.
Tamil textbook PDFs with broken text layers	MED	IndicOCR for extraction; manual check of a sample of chunks.
Data leakage between schools or parents	HIGH	Central scoping helpers, never per-route ad hoc; isolation e2e tests on every route.
Double charging or lost credits	MED	Append-only ledger, idempotent references, transactional updates, concurrency tests.
Poor phone photos from parents	MED	Quality check before charging, capture guidance, retake prompt.
Children's data compliance	MED	Consent records, delete-my-data, file purge, least-privilege access.